Assignment 8

Chapter 7

In the following, $\mathbf{X} = (X_1, X_2, \cdots, X_n)$, $\mathbf{x} = (x_1, x_2, \cdots, x_n)$, and so on.

1. Refer to the discussion in Section 7.4.

   (a) Construct the dependency graph for the random variables involved in the random coding scheme.

   (b) By considering the joint distribution of these random variables, prove the Markov chain in (7.160).

2. Show that the capacity of a DMC with complete feedback cannot be increased by using probabilistic encoding and/or decoding schemes.

4. Consider the channel in Problem 3, Part b).

   (a) Show that the channel capacity is not increased by feedback.

   (b) Devise a coding scheme without feedback that achieves the channel capacity.

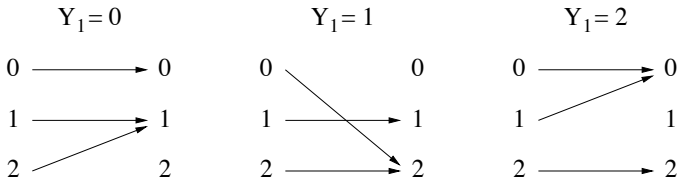
5. In Remark 1 toward the end of Section 7.6, it was mentioned that in the presence of feedback, both the Markov chain $W \to \mathbf{X} \to \mathbf{Y} \to \hat{W}$ and Lemma 7.16 do not hold in general. Give examples to substantiate this remark.

6. Prove that when a DMC is used with complete feedback,

$$\Pr\{Y_i = y_i | \mathbf{X}^i = \mathbf{x}^i, \mathbf{Y}^{i-1} = \mathbf{y}^{i-1}\} = \Pr\{Y_i = y_i | X_i = x_i\}$$

   for all $i \geq 1$. This relation, which is a consequence of the causality of the code, says that given the current input, the current output does not depend on all the past inputs and outputs of the DMC.

12. *Feedback increases capacity* Consider a ternary channel with memory with input/output alphabet $\{0, 1, 2\}$ as follows. At time 1, the output of the channel Y_1 has a uniform distribution on $\{0, 1, 2\}$ and is independent of the input X_1 (i.e., the channel outputs each of the values 0, 1, and 2 with probability $\frac{1}{3}$ regardless of the input). At time 2, the transition from X_2 to Y_2 which depends on the value of Y_1 is depicted below:



   For every two subsequent transmissions, the channel replicates itself independently. So we only need to consider the first two transmissions. In the sequel, we regard this channel as described by a generic discrete channel (with transmission duration equals 2) with two input symbols X_1 and X_2 and two output symbols Y_1 and Y_2, and we will refer to this channel as the *block* channel.

   (a) Determine the capacity this block channel when it is used without feedback. Hint: Use the results in Problems 8 and 11.

(b) Consider the following coding scheme when the block channel is used with feedback. Let the message $W = (W_1, W_2)$ with $\mathcal{W}_1 = \{0, 1, 2\}$ and $\mathcal{W}_2 = \{0, 1\}$. Let W_1 and W_2 be independent, and each of them is distributed uniformly on its alphabet. First, Let $X_1 = W_1$ and transmit X_1 through the channel to obtain Y_1, which is independent of X_1. Then based on the value of Y_1, we determine X_2 as follows:

i. If $Y_1 = 0$, let $X_2 = 0$ if $W_2 = 0$, and let $X_2 = 1$ if $W_2 = 1$.

ii. If $Y_1 = 1$, let $X_2 = 1$ if $W_2 = 0$, and let $X_2 = 2$ if $W_2 = 1$.

iii. If $Y_1 = 2$, let $X_2 = 0$ if $W_2 = 0$, and let $X_2 = 2$ if $W_2 = 1$.

Then transmit X_2 through the channel to obtain Y_2. Based on this coding scheme, show that for the capacity of this block channel can be increased by feedback.